

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Module - I

Practical 1

Roll No.:T007	Name:Subhashree Jena
Class:TYIT	Batch:2
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1. Introduction to Business Analytics:

a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load dataset df =
pd.read_csv("Retail_sales.csv")
```

```
#Display First 5 Records df.head()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
0	Spearsland	52372247	2022-01-01	9	2741.69	20	81	Tanzania	Furniture	Saturday	False
1	Spearsland	52372247	2022-01-02	7	2665.53	0	0	Mauritania	Furniture	Sunday	False
2	Spearsland	52372247	2022-01-03	1	380.79	0	0	Saint Pierre and Miquelon	Furniture	Monday	False
3	Spearsland	52372247	2022-01-04	4	1523.16	0	0	Australia	Furniture	Tuesday	False
4	Spearsland	52372247	2022-01-05	2	761.58	0	0	Swaziland	Furniture	Wednesday	False

```
#Display Last 5 Records
```

```
df.tail()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
29995	Spearsland	50239115	2022-01-25	5	2501.15	0	100	Malawi	Clothing	Tuesday	False
29996	Spearsland	50239115	2022-01-26	3	1500.69	0	0	Sudan	Clothing	Wednesday	False
29997	Spearsland	50239115	2022-01-27	6	3001.38	0	0	South Georgia and the South Sandwich Islands	Clothing	Thursday	False
29998	Spearsland	50239115	2022-01-28	5	2501.15	0	0	Haiti	Clothing	Friday	False
29999	Spearsland	50239115	2022-01-29	3	1425.66	5	190	Mozambique	Clothing	Saturday	False

```
#Display Shape of Dataset df.shape
```

```
(30000, 11)
```

```
#Display Column Names
```

```
df.columns
```

```
Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',  
      'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',  
      'Product Category', 'Day of the Week', 'Holiday Effect'],  
      dtype='str')
```

```
#Display Dataset Information
```

```
df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 30000 entries, 0 to 29999  
Data columns (total 11 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   Store ID              30000 non-null  str  
1   Product ID            30000 non-null  int64  
2   Date                  30000 non-null  str  
3   Units Sold            30000 non-null  int64  
4   Sales Revenue (USD)   30000 non-null  float64  
5   Discount Percentage    30000 non-null  int64  
6   Marketing Spend (USD)  30000 non-null  int64  
7   Store Location         30000 non-null  str  
8   Product Category      30000 non-null  str  
9   Day of the Week       30000 non-null  str  
10  Holiday Effect        30000 non-null  bool  
dtypes: bool(1), float64(1), int64(4), str(5)  
memory usage: 2.3 MB
```

```
#Display Data Types df.dtypes
```

```
Store ID          str  
Product ID       int64  
Date             str  
Units Sold       int64  
Sales Revenue (USD) float64  
Discount Percentage int64  
Marketing Spend (USD) int64  
Store Location    str  
Product Category  str  
Day of the Week   str  
Holiday Effect    bool  
dtype: object
```

```
#Check Missing Values
```

```
df.isnull().sum()
```

```
Store ID      0  
Product ID    0  
Date          0  
Units Sold    0  
Sales Revenue (USD) 0  
Discount Percentage 0  
Marketing Spend (USD) 0  
Store Location 0  
Product Category 0  
Day of the Week 0  
Holiday Effect 0  
dtype: int64
```

```
#Check Duplicate Records
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
#Display Descriptive Statistics
df.describe()
```

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)
count	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000
mean	4.461294e+07	6.161967	2749.509593	2.973833	49.944033
std	2.779759e+07	3.323929	2568.639288	5.974530	64.401655
min	3.636541e+06	0.000000	0.000000	0.000000	0.000000
25%	2.228600e+07	4.000000	882.592500	0.000000	0.000000
50%	4.002449e+07	6.000000	1902.420000	0.000000	1.000000
75%	6.559352e+07	8.000000	3863.920000	0.000000	100.000000
max	9.628253e+07	56.000000	27165.880000	20.000000	199.000000

```
#Count Unique Values df.nunique()
```

```
Store ID          1
Product ID        42
Date              731
Units Sold        32
Sales Revenue (USD) 2545
Discount Percentage 5
Marketing Spend (USD) 200
Store Location     243
Product Category   4
Day of the Week    7
Holiday Effect     2
dtype: int64
```

```
#Display Unique Values of a Particular Column df["Product
Category"].unique()
```

```
<StringArray>
['Furniture', 'Electronics', 'Groceries', 'Clothing']
Length: 4, dtype: str
```

```
#Count Product Categories df["Product
Category"].value_counts()
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```
#Count Store Locations df["Store
Location"].value_counts()
```

```
Store Location
Korea          237
Congo          234
Turks and Caicos Islands 156
Guam           153
Tanzania       151
...
Palestinian Territory 98
Haiti           96
Dominica        95
Ireland         92
Andorra         80
Name: count, Length: 243, dtype: int64
```

```
#Count Holiday Effect df["Holiday  
Effect"].value_counts()
```

Holiday Effect	
False	29836
True	164
Name: count, dtype: int64	

```
#Calculate Average Sales Revenue  
df["Sales Revenue (USD)"].mean()
```

np.float64(2749.509592666667)

```
#Find Maximum Sales Revenue  
df["Sales Revenue (USD)"].max()
```

np.float64(27165.88)

```
#Find Minimum Sales Revenue  
df["Sales Revenue (USD)"].min()
```

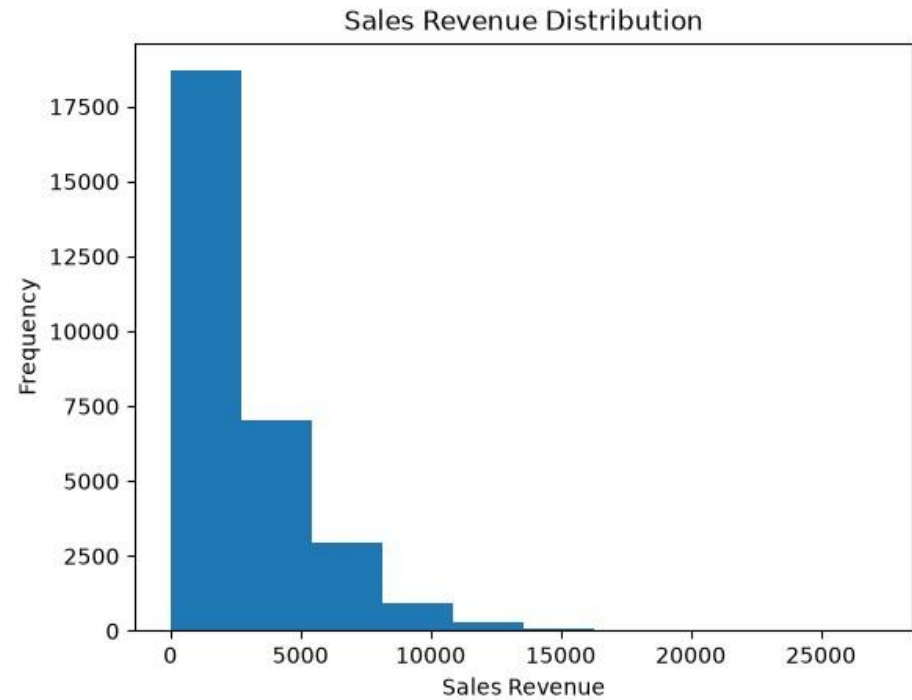
np.float64(0.0)

```
#Correlation Matrix
```

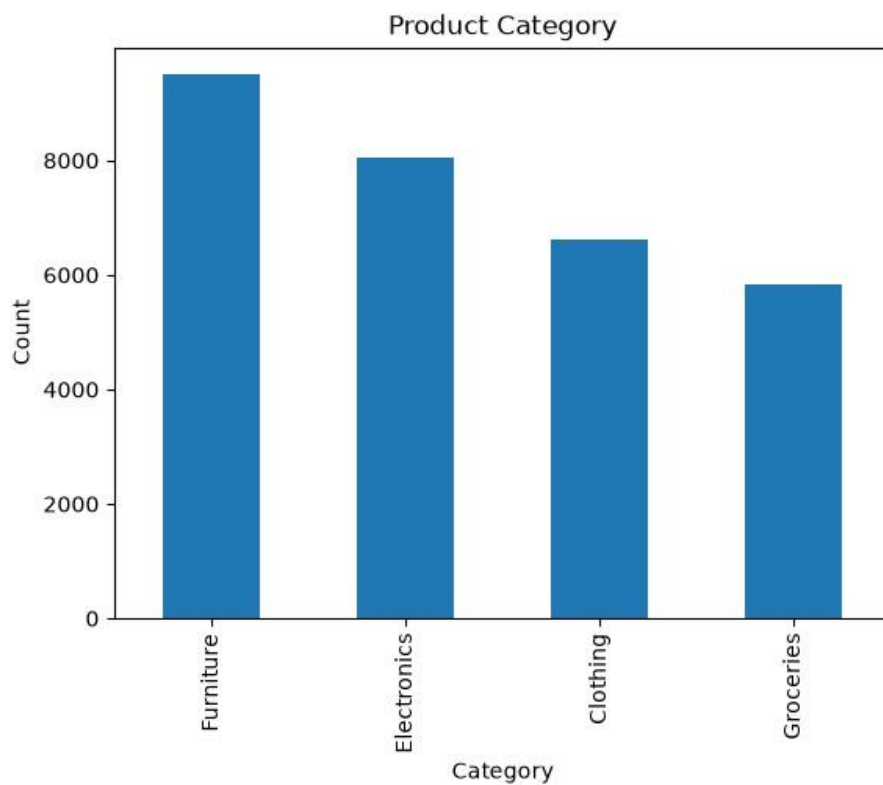
```
df.corr(numeric_only=True)
```

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Holiday Effect
Product ID	1.000000	-0.008570	-0.181934	-0.006891	-0.002496	-0.000015
Units Sold	-0.008570	1.000000	0.615767	-0.001620	-0.004750	0.159324
Sales Revenue (USD)	-0.181934	0.615767	1.000000	-0.065791	-0.002732	0.090681
Discount Percentage	-0.006891	-0.001620	-0.065791	1.000000	-0.000322	0.005848
Marketing Spend (USD)	-0.002496	-0.004750	-0.002732	-0.000322	1.000000	-0.000455
Holiday Effect	-0.000015	0.159324	0.090681	0.005848	-0.000455	1.000000

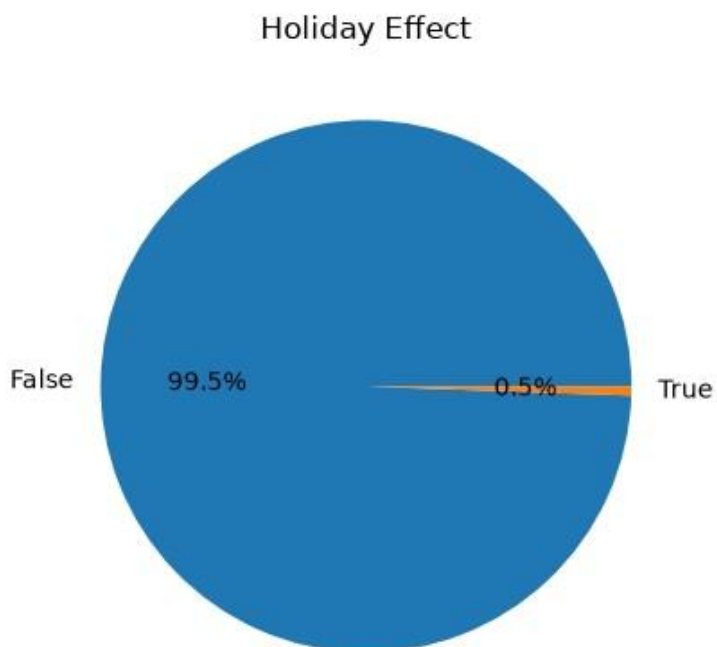
```
#Histogram plt.hist(df["Sales  
Revenue (USD)"]) plt.title("Sales  
Revenue Distribution")  
plt.xlabel("Sales Revenue")  
plt.ylabel("Frequency") plt.show()
```



```
#Bar chart df["Product
Category"].value_counts().plot(kind="bar")
plt.title("Product Category") plt.xlabel("Category")
plt.ylabel("Count") plt.show()
```

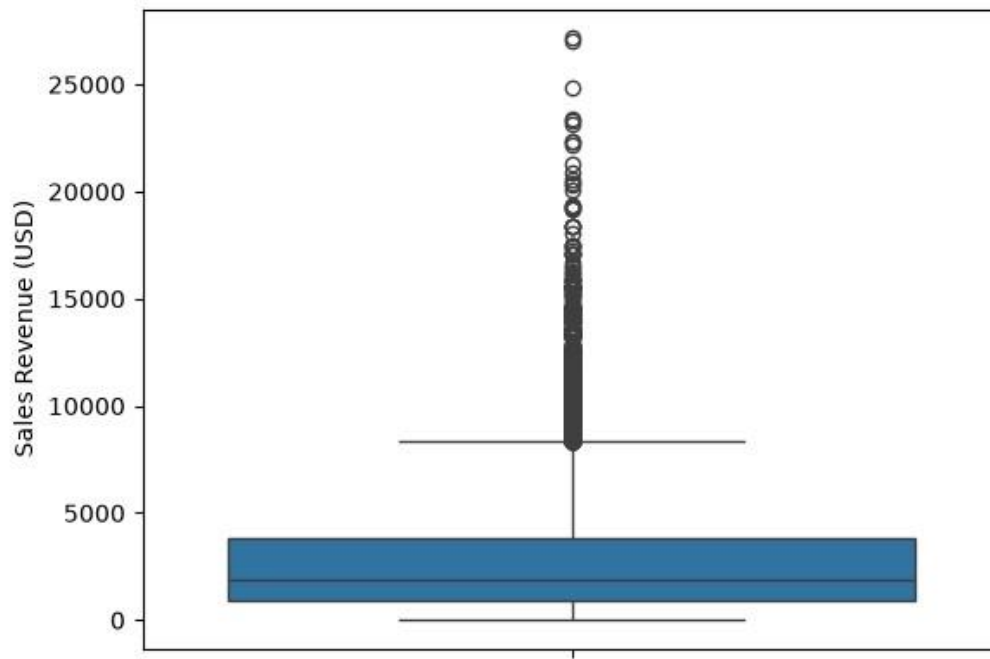


```
#Pie chart df["Holiday Effect"].value_counts().plot(kind="pie",
autopct="%1.1f%%") plt.title("Holiday Effect") plt.ylabel("") plt.show()
```



```
#Box plot sns.boxplot(y=df["Sales
Revenue (USD)"]) plt.title("Sales Revenue
Box Plot") plt.show()
```

Sales Revenue Box Plot



```
#Correlation Heatmap sns.heatmap(df.corr(numeric_only=True),  
annot=True, cmap="Blues") plt.title("Correlation Heatmap")  
plt.show()
```

Correlation Heatmap

